

SHOWCASE

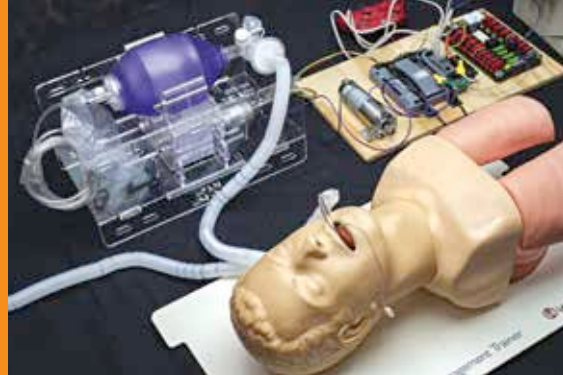
Ultrasona: Ultrasonic cleaning

It's more important now than ever to maintain hygiene. As such, you should be cleaning just about everything you come into contact with. To that end, the Ultrasona: Ultrasonic cleaner is a powerful, small, for-home cleaning device. Like the name suggests, the Ultrasona uses sound, i.e, ultra fast vibrations to clean things. It's not limited to just fabrics and clothes, but a number of things, such as toys, small utensils, and other



knick-knacks. The only limiting factor here is size, the device cannot clean large objects. The Ultrasona uses a 2-part design, one part is the power supply/control box, which is connected via a cord to a transducer. This is the part that vibrates. How it

works is you set a timer and leave it to run, come back in 5-10 minutes and you're done. Pretty simple. You, of course, have to place the Ultrasona and the item you're cleaning in the same container with water. Ultrasonic cleaners aren't new, there have been many commercial cleaners in use already in hospitals and by mechanics for example,. They have, however, never been used for home purposes. Ultrasona hopes to change this by making ultrasonic cleaning available to the masses. The R&D for the device has already been done and they even have supply lines ready. Unfortunately, it won't be shipping until June thanks to the pandemic.



MIT E-Vent Emergency Ventilator

There's no sugarcoating it, the world's currently facing a crisis, and one the biggest issues surrounding it is the lack of ventilators in hospitals. Big companies around the world, such as Dyson, Tesla, Ford and more are working on designing and building ventilators but those will take a while to be officially approved and shipped out.

Fortunately, a group of scientists at MIT have created an emergency ventilator, which is affordable and can easily be made in hospitals using available materials. They've also made the designs to the ventilator available online for easy access. A team of researchers – a combination of scientists, physicians, computer scientists and volunteers – known as E-Vent were the ones who brought back a 10-year-old ventilator project to bring the affordable ventilator design to life. The total cost of the device comes to around \$500, and uses parts that are usually already present in most hospital inventories, such as Ambu resuscitation bags.

The device basically uses a machine to pump the Ambu bag, which is normally manually operated. This means you wouldn't need a person physically standing over the person day and night pumping the bag, and this can be used as a makeshift ventilator until the patient can be moved to a proper ICU ventilator.

Intel Loihi Neuromorphic Chip

Sniffer dogs might soon be out of work, well, at least when it comes to sniffing out dangerous chemicals, drugs and explosives, among other things. Why? Because apparently, AI can now smell.

According to the Daily Mail, researchers at Cornell University and Intel have created a neuromorphic chip called Loihi which allows computers to think like they have biological brains.

A circuit on the chip mirrors the organic circuits one would find in the olfactory sections of a dog's brain. This is basically the part of the brain that controls how dogs process their sense of smell. The Loihi chip can reportedly identify an odour on its first try, and is even able to differentiate it with

other background smells. It can also supposedly detect human smells, and differentiate the smells of healthy and sick humans. It can even identify the illnesses based on the type of smell!

The reason sniffer dogs are as useful as they are is because

not only do they have a powerful sense of smell, they're also able to memorise smells, so the next time they encounter it, they know what it is. For this reason they require plenty of training before heading into the field. With an AI being trained to identify smells, you can be sure that once it's memorised a smell, it will remember for next time. The chip reportedly processes information just like a mammal brain would, using electric signals to process smells.

